Module 20

Entity-Relationship Model/3

Milav Dabgar

www.milav.in

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Module Recap

- ER Diagram for ER Models
- Translation of ER Models to Relational Schema

Module Objectives

- To understand extended features of ER Model
- To discuss various design issues

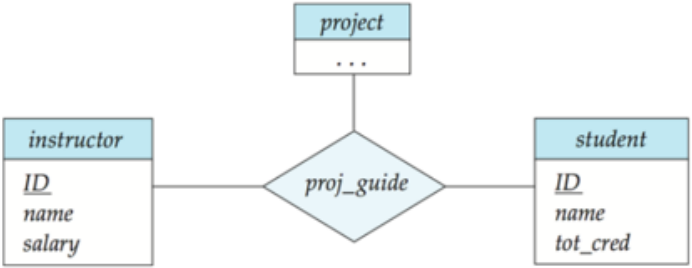
Module Outline

- Extended ER Features
- Design Issues

Extended ER Features

Non-binary Relationship Sets

- Most relationship sets are binary
- There are occasions when it is more convenient to represent relationships as non-binary
- ER Diagram with a Ternary Relationship



Cardinality Constraints on Ternary Relationship

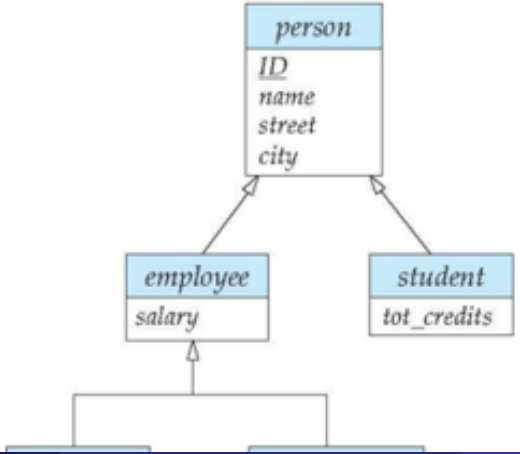
- We allow at most one arrow out of a ternary (or greater degree) relationship to indicate a cardinality constraint
- For example, an arrow from `proj_guide` to `instructor` indicates each student has at most one guide for a project
- If there is more than one arrow, there are two ways of defining the meaning.
- For example, a ternary relationship R between A, B and C with arrows to B and C could mean
  - a) Each A entity is associated with a unique entity from B and C or
  - b) Each pair of entities from (A, B) is associated with a unique C entity, and each pair (A, C) is associated with a unique B
- Each alternative has been used in different formalisms
- To avoid confusion we outlaw more than one arrow

Specialization: ISA

- **Top-down design process**: We designate sub-groupings within an entity set that are distinctive from other entities in the set
- These sub-groupings become lower-level entity sets that have attributes or participate in relationships that do not apply to the higher-level entity set
- Depicted by a triangle component labeled ISA (e.g., instructor 'is a' person)
- **Attribute inheritance**: A lower-level entity set inherits all the attributes and relationship participation of the higher-level entity set to which it is linked

Specialization: ISA (2)

- Overlapping: employee and student
- Disjoint: instructor and secretary
- Total and Partial



Representing Specialization via Schema

- **Method 1:**
  - Form a schema for the higher-level entity
  - Form a schema for each lower-level entity set, include primary key of higher-level entity set and local attributes
- **Drawback**: Getting information about, an employee requires accessing two relations, the one corresponding to the low-level schema and the one corresponding to the high-level schema

schema	**attributes**
person	ID, name, street, city
student	ID, tot_cred
employee	ID, salary

Representing Specialization as Schema (2)

- **Method 2:**
  - Form a schema for each entity set with all local and inherited attributes
- **Drawback**: name, street and city may be stored redundantly for people who are both students and employees

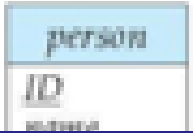
schema	**attributes**
person	ID, name, street, city
student	ID, name, street, city, tot_cred
employee	ID, name, street, city, salary

Generalization

- **Bottom-up design process**: Combine a number of entity sets that share the same features into a higher-level entity set
- Specialization and generalization are simple inversions of each other; they are represented in an ER diagram in the same way
- The terms specialization and generalization are used interchangeably

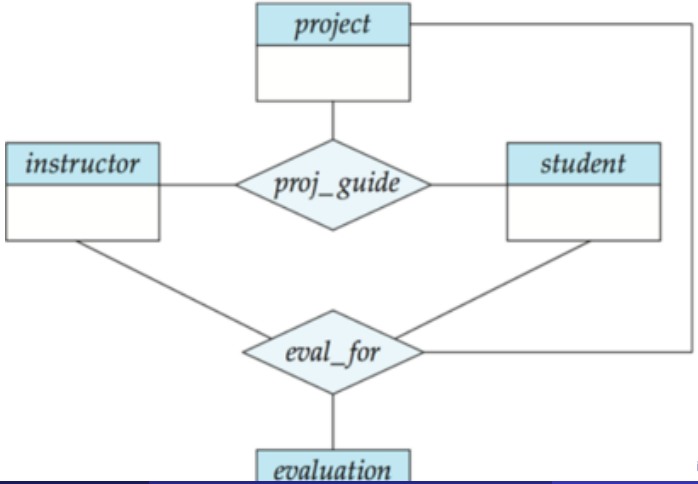
Design Constraints on a Specialization / Generalization

- **Completeness constraint**: Specifies whether or not an entity in the higher-level entity set must belong to at least one of the lower-level entity sets within a generalization
  - **total**: an entity must belong to one of the lower-level entity sets
  - **partial**: an entity need not belong to one of the lower-level entity sets
- Partial generalization is the default. We can specify total generalization in an ER diagram by adding the keyword total in the diagram and drawing a dashed line from the keyword to the corresponding hollow arrow-head to which it applies (for a total generalization), or to the set of hollow arrow-heads to which it applies (for an overlapping generalization).
- The `student` generalization is total. All student entities must be either graduate or undergraduate. Because the higher-level entity set arrived at through generalization is generally composed of only those entities in the lower-level entity sets, the completeness constraint for a generalized higher-level entity set is usually total.



Aggregation

- Consider the ternary relationship `proj_guide`, which we saw earlier
- Suppose we want to record evaluations of a student by a guide on a project

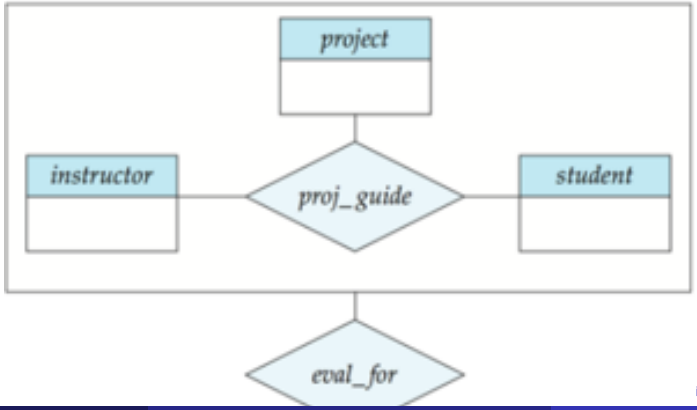


Aggregation (2)

- Relationship sets `eval_for` and `proj_guide` represent overlapping information
  - Every `eval_for` relationship corresponds to a `proj_guide` relationship
  - However, some `proj_guide` relationships may not correspond to any `eval_for` relationships
  - ▷ So we cannot discard the `proj_guide` relationship
- Eliminate this redundancy via aggregation
  - Treat relationship as an abstract entity
  - Allows relationships between relationships
  - Abstraction of relationship into new entity

Aggregation (3)

- Eliminate this redundancy via aggregation without introducing redundancy, the following diagram represents:
- A student is guided by a particular instructor on a particular project
- A student, instructor, project combination may have an associated evaluation



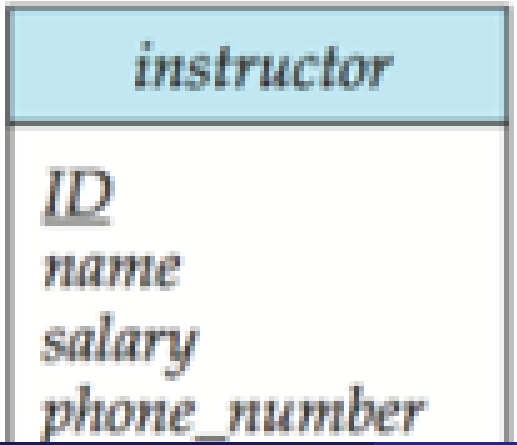
Representing Aggregation via Schema

- To represent aggregation, create a schema containing
  - Primary key of the aggregated relationship,
  - The primary key of the associated entity set
  - Any descriptive attributes
- In our example:
  - The schema `eval_for` is: `eval_for (s_ID, project_id, i_ID, evaluation_id)`
  - The schema `proj_guide` is redundant

Design Issues

Entities vs. Attributes

- Use of entity sets vs. attributes
- Use of phone as an entity allows extra information about phone numbers (plus multiple phone numbers)



Entities vs Relationship Sets

- Use of entity sets vs. relationship sets
  - Possible guideline is to designate a relationship set to describe an action that occurs between entities

- Placement of relationship attributes
  - For example, attribute `date` as attribute of `advisor` or as attribute of `student`

Binary vs Non-Binary Relationships

- Although it is possible to replace any non-binary (n-ary, for $n > 2$) relationship set by a number of distinct binary relationship sets, a n-ary relationship set shows more clearly that several entities participate in a single relationship
- Some relationships that appear to be non-binary may be better represented using binary relationships
  - For example, a ternary relationship `parents`, relating a child to his/her father and mother, is best replaced by two binary relationships, `father` and `mother`
  - ▷ Using two binary relationships allows partial information (e.g., only mother being known)
- But there are some relationships that are naturally non-binary
  - ▷ Example: `proj_guide`

Binary vs Non-Binary Relationships (2): Conversion

- In general, any non-binary relationship can be represented using binary relationships by creating an artificial entity set.
- Replace R between entity sets A, B and C by an entity set E, and three relationship sets:
  1. R_A, relating E and A
  2. R_B, relating E and B
  3. R_C, relating E and C
- Create an identifying attribute for E and add any attributes of R to E
- For each relationship (a_i, b_i, c_i) in R, create
  - a) a new entity e_i in the entity set E
  - b) add (e_i, a_i) to R_A
  - c) add (e_i, b_i) to R_B
  - d) add (e_i, c_i) to R_C

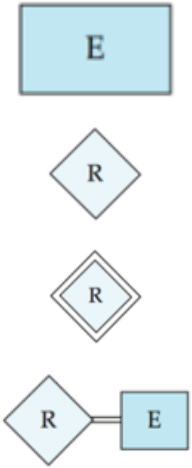
Binary vs Non-Binary Relationships (3): Conversion

- Also need to translate constraints
- Translating all constraints may not be possible
- There may be instances in the translated schema that cannot correspond to any instance of R.
  - ▷ Exercise: add constraints to the relationships R_A, R_B and R_C to ensure that a newly created entity corresponds to exactly one entity in each of entity sets A, B and C
- We can avoid creating an identifying attribute by making E, a weak entity set (described shortly) identified by the three relationship sets

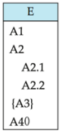
ER Design Decisions

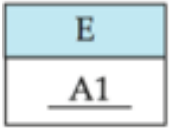
- The use of an attribute or entity set to represent an object
- Whether a real-world concept is best expressed by an entity set or a relationship set
- The use of a ternary relationship versus a pair of binary relationships
- The use of a strong or weak entity set
- The use of specialization/generalization - contributes to modularity in the design
- The use of aggregation - can treat the aggregate entity set as a single unit without concern for the details of its internal structure

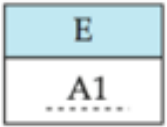
Symbols Used in ER Notation



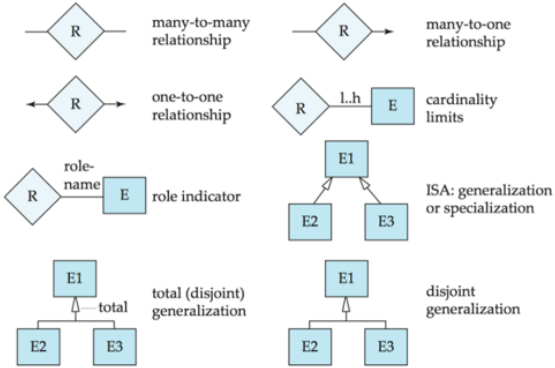
Symbols Used in ER Notation (Cont.)







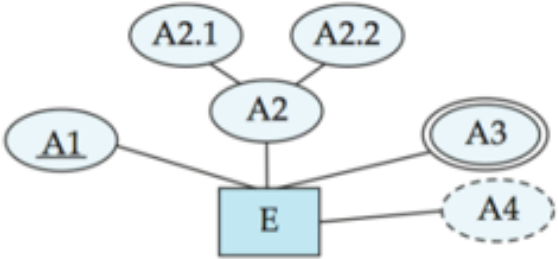
Symbols Used in ER Notation (2)



Symbols Used in ER Notation (3): Alternate

- Chen, IDE1FX, . . .

entity set E with
simple attribute A1,
composite attribute A2,
multivalued attribute A3,
derived attribute A4,
and primary key A1



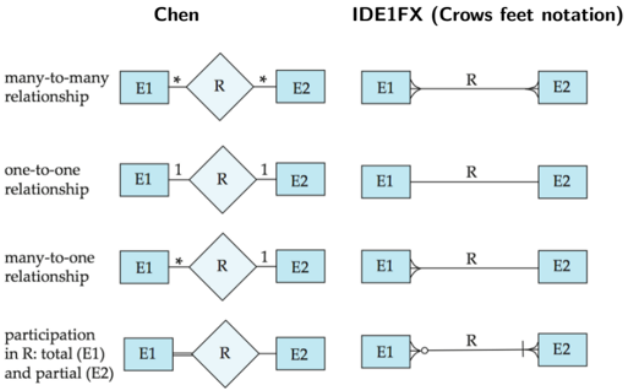
weak entity set

generalization

total
generalization

Symbols Used in ER Notation (4): Alternates

Module 20
Partha Pratim Das
Objectives & Outline
ER Features
Non-binary Relationship
Specialization
Specialization as Schema
Generalization
Aggregation
Design Issues
Entities vs Attributes
Entities vs Relationship
Binary vs Non-Binary
Design Decisions
ER Notation
Module Summary



Module Summary

- Discussed the extended features of ER Model
- Deliberated on various design issues

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